IN5450 MIMO Pulse-Echo Imaging
Project 3 Final Presentation

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Agenda

1. Task 1: Pulse Compression
2. Task 2: Virtual Array
3. Task 3: Delay-And-Sum
4. Task 4: Tapering

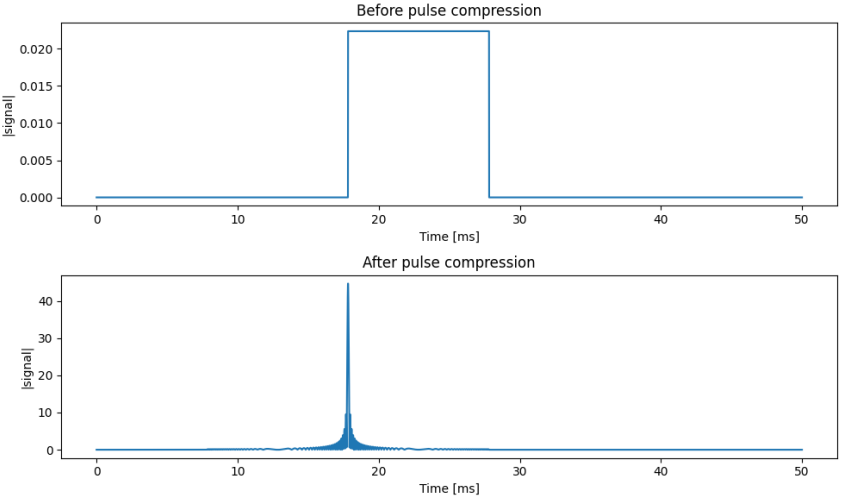
Synthetic Replicas and Compression Pipeline

I used two matched replicas for the coded transmissions:

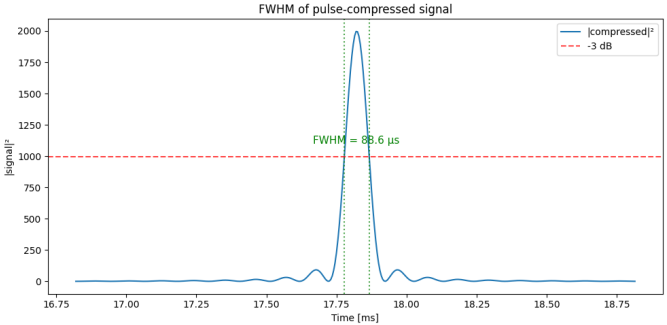
- LFM upchirp for TDMA and one CDMA code channel
- LFM downchirp for the second CDMA code channel

```
alpha = B / Tp
t_p = np.arange(0, Tp, 1 / fs)

s_tx_up = np.exp(1j * 2*np.pi * ((fc - B/2)*t_p + alpha*t_p**2/2))
s_tx_down = np.exp(1j * 2*np.pi * ((fc + B/2)*t_p - alpha*t_p**2/2))

def pulse_compression(data, replica):
    corr = scipy.signal.correlate(data, replica, mode="full")
    return corr[len(replica) - 1:]
```

Single-Channel Result Before and After Compression



Practical Time Resolution (FWHM)

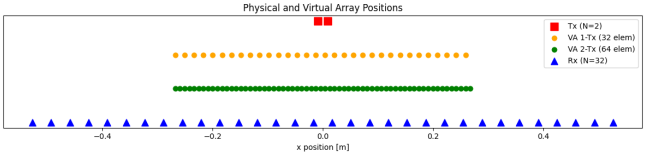


Measured from one compressed timeseries:

- Theoretical: $\Delta\tau = 1/B = 100.00\,\mu s$
- Measured FWHM: $88.60\,\mu s$
- Ratio: 0.886

The compressed peak is much narrower than the original waveform extent, which is exactly what we need for range resolution.

Physical vs Virtual Array (1 Tx and 2 Tx)




```
va_1tx = (tx_pos[0] + rx_pos) / 2
va_2tx = np.array([
    (tx + rx_pos) / 2 for tx in tx_pos
]).flatten()
```

Observation: Adding the second transmitter doubles the virtual channels from 32 to 64 and fills the synthetic aperture more densely. This is the main reason the full TDMA image becomes sharper laterally.

Theoretical Resolution Numbers

Using $\lambda = c/f_c$ and virtual aperture width L_{va}:

$$\Delta\theta = \frac{\lambda}{L_{va}}$$
$$\Delta x = R\,\Delta\theta \quad (R = 4\,m)$$
$$\Delta y = \frac{c}{2B}$$

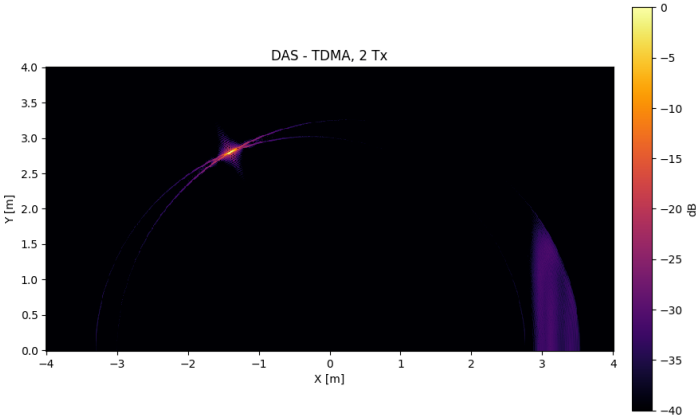
From the notebook output:

- $\lambda = 3.40$ cm, $L_{va} = 0.5355$ m
- Lateral angular resolution: $\Delta\theta = 0.0635$ rad (3.64°)
- Lateral resolution at 4 m: $\Delta x = 25.40$ cm
- Axial resolution: $\Delta y = 1.70$ cm

DAS Implementation (True Bistatic, Nearest Neighbor)

```
def vectorized_das(data, grid_x, grid_y, tx_positions, rx_positions, fs, c):
    GX, GY = np.meshgrid(grid_x, grid_y)
    image = np.zeros((len(grid_y), len(grid_x)), dtype=complex)
    for itx in range(n_tx):
        R_tx = np.sqrt((tx_positions[itx] - GX)**2 + GY**2)
        for irx in range(n_rx):
            R_rx = np.sqrt((rx_positions[irx] - GX)**2 + GY**2)
            R_total = R_tx + R_rx
            idx_map = np.round(R_total / c * fs).astype(int)
            valid = (idx_map >= 0) & (idx_map < Nt_data)
            idx_safe = np.clip(idx_map, 0, Nt_data - 1)
            pixel_values = data[idx_safe, irx, itx]
            pixel_values[~valid] = 0.0
            image += pixel_values
    return image
```

Full TDMA Image in dB (2 Tx)

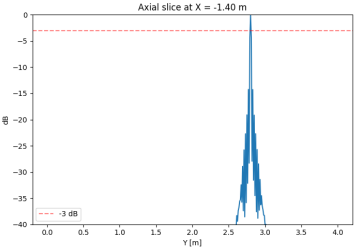


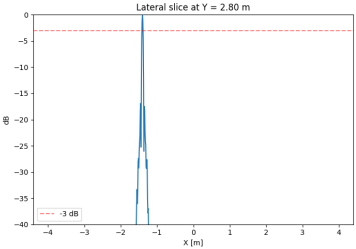
Peak reflector from the reconstructed image:

- $x = -1.400\ m$
- $y = 2.800\ m$

This location is used as reference for the slice-based resolution estimate.

Measured vs Theoretical Resolution





Measured (from -3 dB width):

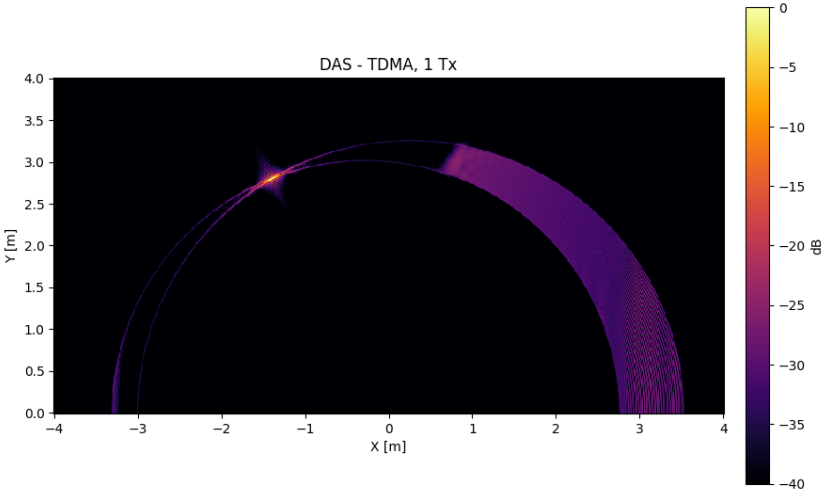
- Axial FWHM: 2.00 cm
- Lateral FWHM: 4.00 cm

Theory from Task 2:

- Axial: 1.70 cm
- Lateral: 25.40 cm

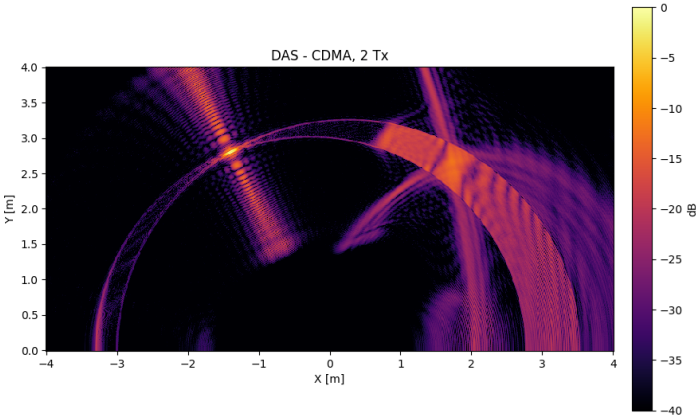
The practical lateral width is much smaller around this point target than the simple aperture formula predicts.

Single-Transmit TDMA vs Full TDMA



With only one transmitter, the reflector is still visible, but the right-side arc artifact and broader energy spread are clearly stronger than in the full 2-Tx case.

CDMA Image and Cross-Talk Pollution



CDMA is reconstructed by matched filtering with both chirp replicas and summing DAS outputs.
The bright streaks and structured haze come from imperfect code separation and residual cross-correlation leakage between upchirp and downchirp channels.
Pollution level is typically in the −20 to −30 dB range around the main target in this map.

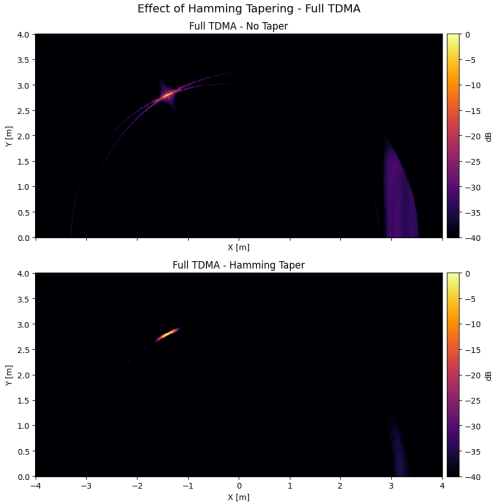
Hamming Tapering on Array and Waveform

```
w_rx = np.hamming(Nrx)
w_tx = np.hamming(len(s_tx_up))
s_tx_up_tapered = s_tx_up * w_tx

# Tapered pulse compression + weighted DAS
tdma_compressed_tapered = ...
image_tapered = vectorized_das_weighted(
    tdma_compressed_tapered, grid_x, grid_y,
    tx_pos, rx_pos, fs, c, rx_weights=w_rx
)
```

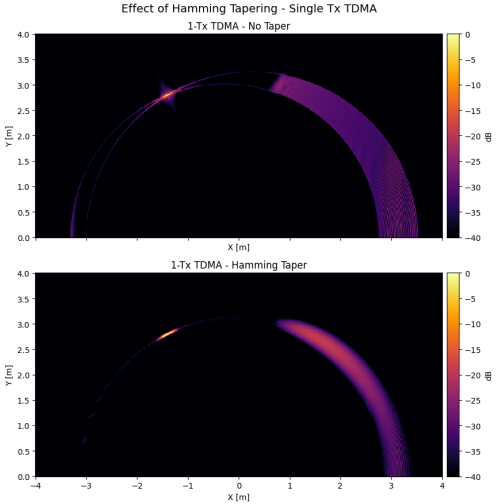
Measured trade-off from notebook output:

- PSL improves from -18.0 dB to -32.7 dB
- Mainlobe widens: axial $2.00 \rightarrow 4.00$ cm, lateral $4.00 \rightarrow 6.00$ cm

Original vs Hamming (Full TDMA)



Original vs Hamming (Single-Tx TDMA)



Summary

- Pulse compression gave the expected narrow peak and practical $\Delta\tau$ close to theory.
- Using both transmitters expanded the virtual aperture and improved image sharpness.
- True bistatic DAS gave the cleanest TDMA image. CDMA showed visible cross-talk artifacts.
- Hamming tapering reduced sidelobes significantly, with loss in mainlobe resolution.